

Use of Scrum Methodology in System Development

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Abstract— The Scrum approach has become one of the most important frameworks within the Agile methodology. This analysis paper focuses on the Scrum methodology, analyzing its specified roles, change policies, delivery cycles, task handling and also presents a comparative view with Kanban, another important Agile method. The Scrum approach encourages collaboration among teams and iterative improvement by using organized sprints and defined roles. Meanwhile, Kanban offers flexibility through a continuous process approach. Moreover, this paper will discuss the advantages and disadvantages of the Scrum method in various working environments. Lastly, the paper is then wrapped up with the conclusion statement.

Keywords— Scrum, Kanban, Agile Methodology, sprint

I. INTRODUCTION

Software development project management has changed rapidly as a result of agile approaches, which have introduced frameworks like Scrum and Kanban that significantly enhance teamwork and flexibility [1]. In order to lead a project through iterative sprints, Scrum offers an organized yet flexible approach that includes roles such as Scrum Master, Product Owner and Development Team [2]. These sprints, which are short, restricted in time periods, enable teams to divide large projects into tiny tasks. Moreover, it encourages iterative development and frequent feedback. This method not only encourages constant improvement, but it also assures

that the final product is in line with the client's expectations and market demands.

Kanban, on the other hand, provides a continuous delivery structure with a focus on visual management and adaptability. Unlike Scrum, Kanban does not require clear role classification [2]. This allows teams to make adjustments as needed without having to stick to a fixed timeline. This method is especially useful for projects with changing requirements because it enables teams to respond rapidly to new information and any changing demands.

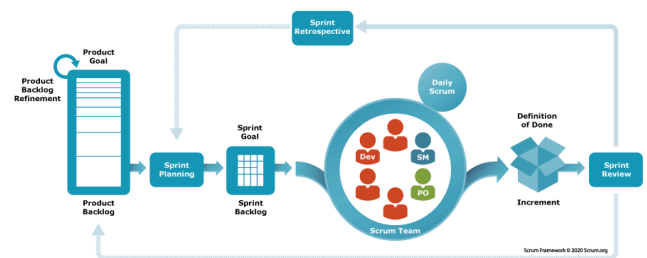


Figure 1.1 Scrum framework

II. COMPARATIVE ANALYSIS

In the context of software development, several methods serve as important frameworks that determine how teams organize their tasks and develop products. Scrum is well-known for its structured approach and defined roles, while Kanban is adaptable and emphasizes continuous workflow and visual management [3]. Understanding the differences between Scrum and Kanban can assist teams in selecting the most appropriate approach [4]. Some of

the differences of Scrum and Kanban methods are briefly described as under:

TABLE I
COMPARATIVE BETWEEN SCRUM AND KANBAN

Agile Method	Scrum	Kanban
Roles of Employee	Scrum method has specified roles, such as Scrum Master, Product Owner and Development Team.	Kanban does not define specific roles.
Change Policy	In Scrum, changes are usually restricted after a sprint begins.	Kanban enables changes to be implemented at any point.
Delivery Cycle	Scrum has a specific delivery cycle with sprints that range between 1 to 4 weeks.	Kanban’s delivery cycle is continuous.
Work Limit	Scrum manages the workload by assigning a specific limit for each sprint.	Kanban’s work limit uses visual management, which is known as Work In Progress (WIP).

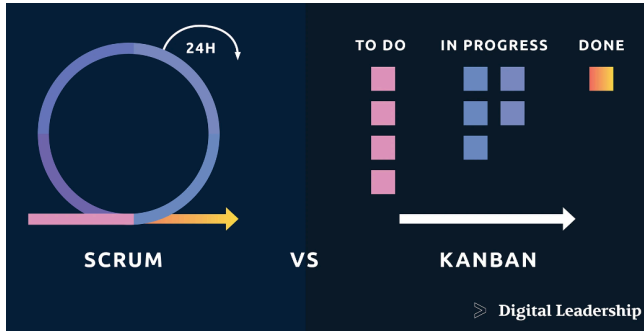


Figure 2.1 Timeline difference between Scrum and Kanban

III. ADVANTAGES

Scrum has many advantages that make it ideal for handling complex projects, especially in dynamic environments where flexibility, teamwork and iterative development are key to success.

Scrum prioritizes the most valuable tasks and divides work into concise and focused sprints to speed up project delivery [5]. They typically run targeted sprints that usually last two to four weeks where the team focuses on completing a small number of high-priority projects. This sprint structure allows the team to handle work in small and manageable increments to reduce the possibility that the entire project scope would overwhelm them. The team will make sure that every cycle produces significant progress and meets the highest priority needs by focusing only on the most valuable tasks during each sprint. This strategy not only improves concentration but also enables stakeholders to observe measurable outcomes early in the project and make necessary adjustments.

Next, Scrum fosters enhanced collaboration by promoting Scrum values into every aspect of the process, creating a highly cooperative environment [6]. Smooth teamwork is made possible by the defined objects with commitments, the organized Scrum events and the unique accountabilities associated with each Scrum role. The majority of duties can be completed by several team members because Scrum Teams are meant to be multipurpose. When it comes to more complicated tasks, like creating solutions and resolving difficult issues, two or three developers working together frequently provide better, quicker results than working alone. This method not only increases productivity but also makes use of the team's variety of skills to provide better outcomes.

Additionally, by making it easier to incorporate changes into ongoing work, Scrum adds flexibility to project management [7]. Scrum also allows the scope to remain flexible while keeping

consistent time and cost limitations because of the structure, each sprint has a predetermined duration and set of resources. The team can adjust priorities inside each sprint thanks to the predictable structure this offers, all without compromising the project's overall budget or schedule. This adaptability enables teams to react to changing client demands and fresh perspectives, producing value gradually without raising prices or delaying deadlines. Ultimately, customer satisfaction and project efficiency will be increased.

In Scrum structure, the team shows their progress at the conclusion of each sprint in order to get timely feedback and make necessary adjustments without having to redo a lot of work. Stand-ups meeting keep everyone on the same page and resolve problems in real time. Scrum makes it possible to produce high-quality solutions more quickly and effectively by providing a potentially deployable product increment at the conclusion of each sprint.

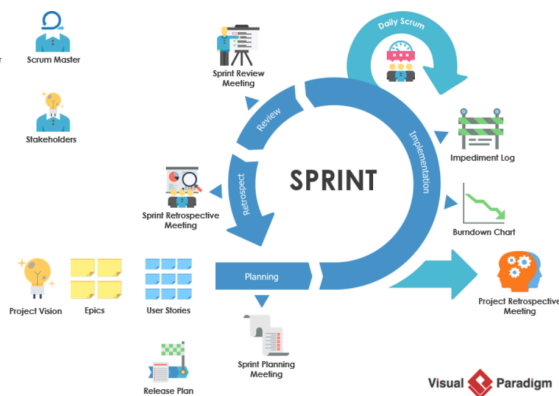


Figure 3.1 Key element of Scrum framework

IV. DISADVANTAGES

Scrum offers many benefits, but it also has several drawbacks that can cause problems for teams and organizations especially if the framework is not completely understood or applied correctly.

Small, targeted teams which ideally consist of 3 to 10 people are essential to Scrum in order to optimize cooperation and sustain efficient communication [8]. This team size makes sure close cooperation and quick decision making are necessary. However, some companies might find it difficult to

modify their personnel structure to fit this arrangement. It can be logistically challenging and sometimes taxing to divide up huge departments into several small, independent teams. Furthermore, companies with specialized positions can find it difficult to divide up talents among teams fairly, which could affect cohesiveness and production.

Integrating Scrum with traditional project management can be challenging, particularly for projects that call for meticulous prior preparation and tight predictability [9]. Although Scrum is excellent at flexibility and iterative development, it lacks the long-term planning focus of methodology like Waterfall, which might be more appropriate for projects with set milestones and consistent needs. Scrum's short-term, flexible planning may be difficult for organizations that use a linear, rigid method to integrate with their current procedures.

Finally, team members can be under significant stress due to Scrum's fast-paced, iterative approach and frequent deadlines [10]. The need to complete a specific amount of work during each sprint cycle often creates a high-intensity environment where team members must work quickly to meet sprint goals. Regular participation is necessary for daily stand-ups, sprint reviews, and retrospective meetings, but they don't provide much opportunity for reflection or recovery. Stress and even burnout may result from this continuous workload, particularly if the team is understaffed or sprints are not adequately managed. Additionally, team members can take on extra responsibilities to ensure that goals are met because Scrum encourages self-organization, which would add to their workload.

V. CONCLUSIONS

In summary, Scrum promotes collaboration, flexibility, and incremental development, providing a useful framework for handling complex projects. Thanks to organized sprints and assigned roles, teams are able to produce high-quality work while accepting changes and taking feedback into account. Scrum has benefits for dynamic, adaptable settings, but it also

has drawbacks, especially for businesses that require predictability, larger team sizes, or interface with more traditional project management techniques. Overall, Scrum offers tremendous benefits to projects that thrive on iterative development and close collaboration, making it a valuable approach for teams that can adapt to their unique needs.

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